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Automated generation and semantic segmentation of roof orthophoto for digital twin -based monitoring of slated roofs

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ABSTRACT

Roofs are one of the building elements most exposed to environment-induced deterioration, and their deterioration can rapidly result in potential safety hazards to occupants or even buildings' failure. In pursuit of cost-effective, efficient and pro-active building roof monitoring, this paper presents a computing pipeline for: 1. Automated roof orthophoto generation that uses as input the 3D reconstruction from structure-from-motion photogrammetry and the building's Digital Twin (DT) model, and outputs occlusion-free panel orthophotos with metric resolution. 2. Automated orthophoto semantic segmentation using deep learning to extract visible roof sub-components from the orthophotos. Various segmentation models are tested and contrasted. The approaches are validated with three real case studies of traditional buildings with slated pitched roofs. The proposed approach employs a generic pipeline which could be applied to any element of the building fabric.

1. Introduction

As a key component of building infrastructure, the roof is highly exposed to environmental elements. Elevated temperatures, moisture, and material soiling can lead to staining, weathering, and damage to various materials [1]. Over time, these factors contribute to increased roof fragility, component failure, and load redistribution, posing additional risks to the building and its occupants [2]. For instance, significant exposure to weather often leads to deterioration in the slated areas of building roofs [3]. Moreover, extreme climate events such as high summer temperatures, heavy winter precipitation, unpredictable winds, and intense storms are exacerbating the deterioration of buildings. In the case of traditional buildings, proactive conservation measures to prevent failures have become crucial [4]. Consequently, routine monitoring of building roofs is essential to detect and address defects promptly.

Visual inspection is a widely established method for inspecting and diagnosing building roofs, both flat and pitched [5,6]. However, conventional methods heavily depend on manual effort for both observation and analysis: surveyors must access roofs physically, requiring them to climb and walk around, which demands specific skills and experience. Typically, scaffolding is necessary for safe access, significantly increasing the time, cost, and safety risks associated with the inspection process. Overall, the traditional manual roof monitoring approach is costly, inefficient, and fraught with safety hazards. In contrast, the use of Unmanned Aerial Vehicles (UAVs) for capturing digital images is increasingly popular, significantly reducing the costs and safety risks associated with initial surveys [7,8].

While the advancements in efficient and safe visual data acquisition are widely adopted in practice, the analysis of the acquired images often remains a manual process, with results typically documented in an unstructured format within text documents [6]. For historic buildings, it is common to include manually annotated orthophotos of façades (which cover pitched roof panels) to indicate deteriorations and defects (refer to Fig. 1). Orthophotos offer the advantage of providing uniform scale across the entire image, facilitating the measurability of all details. This uniformity makes it easier to compare deterioration across different sections of the orthophoto, thereby aiding in quantifying maintenance and repair needs [9]. However, generating orthophotos for certain architectural elements, especially roofs, can be problematic due to potential occlusions by chimneys or other structures.

In the broader Facility Management (FM) industry, Digital Twins of building assets — typically derived from Building Information Modelling (BIM) models — are proposed to record life cycle information of assets and support asset managers make decisions related to operation and maintenance, potentially aided by Artificial Intelligence (AI) [11]. However, this concept remains largely aspirational, with actual implementations still in their early stages and primarily considered for new buildings, which benefit from modern design and construction practices

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Fig. 1. Rectified image of façade with overlays showing stone decay codes and repair needs. Note: the hatched doorway area was measured on site for image scaling as was the average block dimensions (British Geological Survey image P606737 ©UKRI 2005).
Source: This image is reproduced with agreement from [10].

incorporating BIM models. In contrast, exploring the feasibility and potential challenges of applying such practices to traditional buildings, particularly in the context of roof monitoring, has received limited attention [12,13].

This paper aims to contribute to the development of DT-based solutions for the safe and efficient monitoring of roofs. The paper more specifically focuses on traditional slated roofs that are prominent in historic buildings but still common in modern construction. First, we propose a method for generating high-resolution vertical orthophotos for pitched building roofs using an existing building DT. This method aims to produce images that are free from occlusions caused by other building elements and exhibit reduced blur. Secondly, we develop a deep learning-based approach for semantically segmenting the orthophotos into their constituent sub-elements, such as slates, leadwork, and others. This segmentation is crucial as it allows subsequent deterioration and defect detection tools to operate optimally, being trained and applied specifically to the materials they are designed to assess.

The remaining of this paper is organised as follows. Section 2 reviews the literature and state of the art on: roof inspection, the applications of orthophoto and photogrammetry, and image semantic segmentation techniques. Section 3 gives an overview of the proposed pipeline, with the methods for its two successive stages of orthophoto generation and orthophoto semantic segmentation detailed in Sections 4 and 6, respectively. Sections 5 and 7 report the experimental results obtained for those two stages, respectively. Finally, Section 10 concludes the work, reviewing its strengths and limitations and offering some outlooks for future work.

This paper builds on the two conference papers, [14,15], that introduced the two stages of the method presented here, i.e. orthophoto generation and orthophoto semantic segmentation. This paper revisits the presentation of the methods (in particular the orthophoto generation one) and extends their validations with additional case studies.

2. Related works

2.1. Digital twinning technologies for building management and monitoring

DT technology can be used for many purposes in the built environment: A BIM-based DT is used as a database, which links real-time collected data (e.g., occupancy count) and static data (e.g., maintenance history and technical documentation) to the 3D representation of facility [16]. These links help ease information retrieval for various building monitoring and maintenance applications. For example, Seghezzi et al. [17] propose a Building Management System (BMS) based on an occupancy-oriented DT to detect actual room occupancy, for optimising space management, organisation, and cleaning activities. Another area of active research in the DT domain is the creation of DTs for existing buildings. For example, Bassier and Vergauwen [18] present a novel method to automatically reconstruct BIM wall objects and their topology, Lin et al. [19] use the 3D reconstructed data of elevated structures to realise defect detection within an end-to-end robotic bridge inspection system, and Armeni et al. [20] propose a hierarchical, detection-based approach for semantically parsing the point cloud data of buildings.

In the context of historic buildings, Rocha et al. [21] employ a manual modelling workflow based on photogrammetry and laser scanning data to develop a BIM model of a historic house in Lisbon for conservation and restoration.

Within the context of Digital Twinning for building fabric monitoring, there is great interest in automating data acquisition and processing in order to efficiently, robustly and safely detect and monitor defects, and support computer-aided maintenance decision-making [22]. Regarding defect detection, Valero et al. [23] use machine learning to detect defects in 3D point clouds of masonry walls using machine learning, while Idjaton et al. [24] detect spalling zones in orthophotos of limestone walls using a deep neural network. And Loverdos and Sarhosis [25] apply deep learning to a large brickwork masonry walls image dataset to realise automated brick segmentation and crack detection. Regarding roof image analysis, work seems to have mainly be reported for the monitoring of flat roofs. Yudin et al. [26] use aerial images for flat roof defect segmentation (hollows, swelling, folds, patches and breaks), based on several deep neural networks.

To our knowledge, no work has been reported to date that focuses on pitched slated roofs. In comparison to those other building fabric elements, historic pitched slated roofs have a more complex composition with multiple sub-components (slate, leadwork, tower, etc.), which brings challenges to defect detection. To realise DT-based slated roof panel condition monitoring, it is thus necessary to distinguish the sub-components inside the roof panel orthophoto, so that specifically-built defect detection solutions can be applied to each sub-component with uniform materials.

2.2. Photogrammetry in AEC-FM sector

As a popular capturing reality method, digital structure-from-motion photogrammetry has been an active topic for generating 3D models of buildings due to its high level of automation and the fact that it can generate 3D data at low cost [27]. Thanks to the increased availability of UAVs, *UAV-based structure-from-motion photogrammetry* has been applied in several contexts, such as archaeology, disaster monitoring, construction monitoring, building surveying, or land mapping [28]. Due to the feature of non-destruction and remote sensing, this technology is particularly suitable for roof data capturing.



Fig. 2. Orthophoto (commonly called 'elevation') of side of a residential house.

Structure-from-motion photogrammetry can support various use cases in the AEC-FM sector. An important one is the creation of semantically-rich 3D models, or Digital Twins (DTs), of existing buildings or infrastructure assets, commonly referred to as 'Scan-to-BIM' [29–31]. In particular, Golparvar-Fard et al. [32] have focused on automatically generating as-built models from unstructured collection of construction images by digital photography (Scan-to-BIM by photogrammetry). Civil infrastructure elements and construction operations can be monitored by aligning the photogrammetric reconstructed model to the as-planned BIM model (Scan-vs-BIM) [33]. During maintenance, 'Scan-vs-BIM' [34] can be applied, by comparing the as-designed model and 3D data, to identify the objects and any change (e.g. structural displacements).

2.3. Orthophotos in building monitoring

As a photogrammetric map with a uniform dimensional scale, an *orthophoto* corrects the distortion and relief, and its generation is achieved by an orthographic projection on vertical planes [35,36]. Orthophotos are convenient and are widely used in practice for monitoring buildings for actual surface measurement and analysis, in particular façade elevations. In recent research, high-resolution orthophotos of façades have been used as input data for automated detection of façade defects [23,24,37].

However, different from building façades, roofs (especially pitched roofs) have a more complex composition and are often partially occluded by other building elements, such as chimneys, dormers, aerials, satellite dishes, etc [38]. As a result, building façade elevation orthophotos may not enable a full view of roofs, which can cause difficulty in monitoring. Fig. 2 shows an example of façade elevation orthophoto commonly generated for a building. One can see that, while the stone façade(s) appear(s) fully, the middle roof panel is occluded by the chimneys, pipes, etc. These problems challenge the generation of clear and complete orthophotos (without occlusion and with limited blur), and ultimately the development of automatic defect detection models, as the models need to be agnostic to different image acquisition situations (e.g. angle and distance).

Moreover, roof panels typically consist of sub-components or parts made from a variety of materials, such as slate tiles and leadwork. For the effective application of defect detection methods, such as those previously presented in the context of masonry, it becomes essential to first segment the orthophotos into these sub-components. This segmentation allows for the isolation of target areas, particularly slate tiles which cover the largest areas and are most susceptible to frequent deterioration—thereby facilitating more precise and effective analysis.

2.4. Semantic segmentation and its use in construction informatics

2.4.1. Semantic segmentation

Semantic segmentation was developed decades ago and has been developed into a well-established tool in Computer Vision with its powerful pixel-level segmentation ability. This technique is applied to different types of data, including 2D image and video, and 3D point cloud data. In construction informatics research dealing with 2D and 3D data (e.g. Scan-to-BIM), semantic segmentation is increasingly regarded as an essential step after data collection, to provide further information useful to support subsequent tasks such as object detection [39]. This technique has been applied to different types of data, including: (1) 2D image of indoor scene [40] and aerial images of different architecture [41]; (2) 3D point cloud of building interiors [39], plumbing and structural components [42], 4D point cloud (3D point cloud videos with temporal information) captured by auto-driving cars and robot [43].

Specifically in its application to 2D images, semantic segmentation enables assigning a class label to each pixel of an image, and such pixel-level semantic information can help make judgements or be leveraged by other tasks [44,45]. Traditional machine learning methods, such as object contour detection, recognition, simple filtering and other image processing work, can realise segmentation, especially in some contexts where the colour, texture and other features are uniform within the same object, e.g. in some medical science studies [46]. These machine learning-based methods rely on the careful design of feature extractors by humans, but they have the advantage of lower requirements in terms of computational power and data quantity.

In comparison, deep learning models aim to learn those features from the data itself and commonly outperform machine learning models if they are trained with sufficiently large and representative datasets. Different deep architectures have been introduced, Convolutional Neural Networks (CNNs) being prominent. As a result of active research in this area, the performance of semantic segmentation models has greatly improved, not only in accuracy but also in speed [47]. different CNN-based deep learning models have been proposed for 2D image data segmentation. *DeepLabV3*, a widely used system developed by Google [48], can handle the problem of segmenting objects at multiple scales with cascaded module and Atrous Spatial Pyramid Pooling(ASPP). Compared to *DeepLabV3*, *PointRend* demonstrates better performance (higher mIoU value), by extracting the point features made up of fine-grained features and coarse prediction [49]. Specifically, *PointRend* handles better the problems of smooth regions and blurry contours from the bilinear sampling process. Thanks to the recognisable features, objects can be easily detected and classified, but the precision of contour detection is low.

In recent years, Vision Transformer (ViT) has achieved performance superior to the commonly used CNNs, by splitting each image into patches. Due to their outstanding performance in image classification, these models have also been explored for semantic segmentation: *Segmenter* can capture global interaction between elements of a scene using transformer, rather than the traditional convolutions, which would easily cause information loss [50]. *SegFormer* reaches better performance and efficiency by re-designing the Transformer encoder and uses a simple multilayer perceptron (MLP) decoder [51]. However, all authors also point out that ViT relies heavily on large-scale datasets to achieve good performances [52].

Within these main classes of deep learning models, different variants can be created by modifying different components, such as the loss function. *Focal Loss* is a commonly used loss function for dealing with class imbalance and putting more focus on the hard and misclassified examples, by multiplying each class loss with a weighting factor [53]. *Dice Loss* can deal with the imbalance problem between foreground and background, by giving more importance to the foreground than the background, thus making the model region-related [54]. These two loss functions alongside the default *Cross-Entropy Loss* can be paired and used together in improving model training [55].

Besides, label correction is a common practice in semantic segmentation, particularly in the task of 2D image segmentation. Label correction involves refining and updating the annotations of training data to improve the quality and accuracy of segmentation models. This process addresses issues such as annotation errors, inconsistencies, and

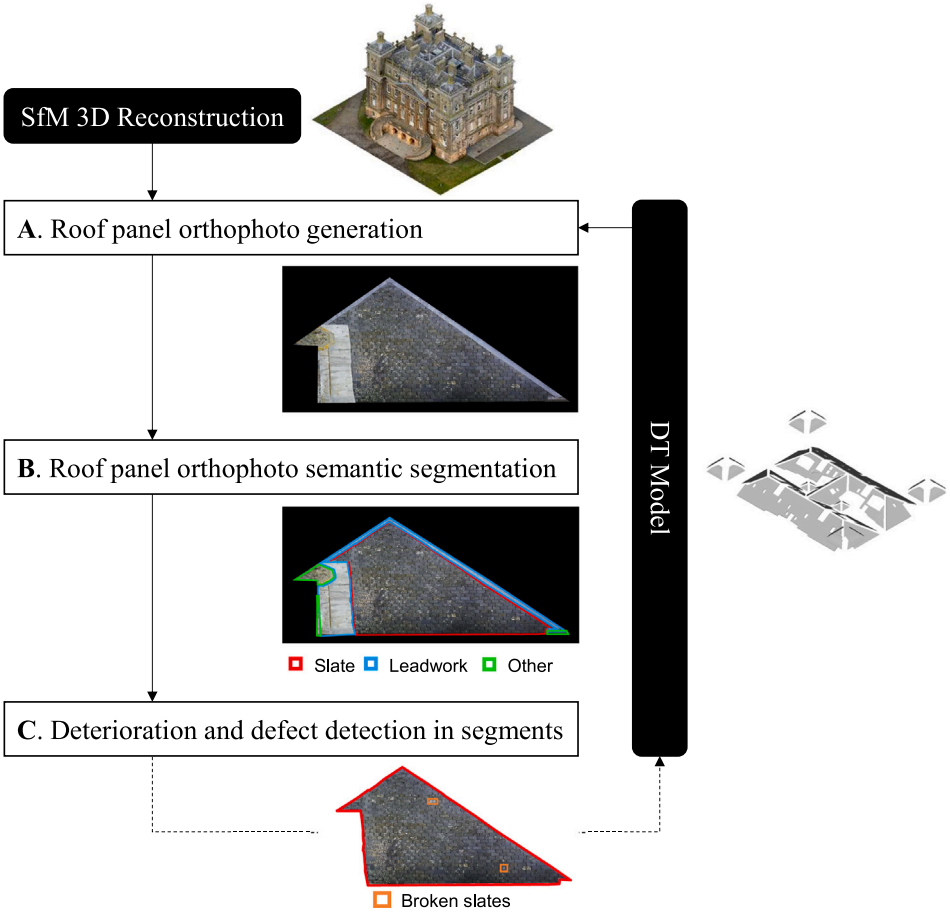


Fig. 3. Overview of the data processing pipeline for monitoring slated pitched building roofs.

ambiguities, which are prevalent in large datasets. Wang and Yushkevich [56] developed a multi-atlas segmentation method with joint label fusion and corrective learning, which combines segmentations from multiple atlases and corrects label errors during training. Hesse and Namburete [57] focused on improving U-Net segmentation through active contour-based label correction, which corrects imperfect ground-truth labels by aligning them with object boundaries, thus enhancing boundary predictions in medical ultrasound imaging. Ibrahim et al. [58] introduced a semi-supervised framework with self-correcting networks, which uses a small set of fully supervised images and a larger weak set with only bounding box labels. Their self-correction module iteratively improves the generated labels during training, significantly reducing the annotation effort while maintaining high performance. Similarly, Mendel et al. [59] proposed an Error-Correcting Supervision approach, where a secondary correction network assists the primary segmentation model by generating pseudo-labels for unlabelled data during training, resulting in consistent improvements over supervised baselines. Integrating these advanced label correction techniques into the training pipeline not only enhances performance metrics but also contributes to the generalisation ability of the models, making them more effective in real-world applications. The semantic segmentation method employed herein also applies a contour-based label correction, using the prior knowledge contained in the BIM model that the segmented images are aligned with.

2.4.2. Semantic segmentation in construction informatics

Semantic segmentation has been actively explored in construction informatics using different models, now principally deep learning models. Examples include: equipment and worker segmentation in images for visual understanding on construction sites [60], building interior segmentation of point clouds [20], material and damage (e.g., cracks, spalling, and corrosion) segmentation in images [61,62] and point cloud [63,64]. In the context of historic building monitoring, semantic segmentation has also been used in masonry wall semantic segmentation [65], crack segmentation in masonry wall images [66]. However, there is seemingly no specific research dealing with semantic segmentation of data (images or point clouds) from other building fabric elements like roofs.

2.5. Contribution above state of the art

This study develops an automated pipeline for monitoring slated pitched building roofs. It produces an occlusion-free orthophoto for each roof panel, which is then semantically segmented into individual sub-components including ‘slate’, ‘lead’, and ‘other’. This method, following a ‘Scan-vs-DT’ approach, facilitates the use of deep learning models for detecting deteriorations and defects in each sub-component area. Ultimately, this allows for the comprehensive documentation of all deteriorations and defects, along with their progression, in the roof DT (logbook). While tailored for pitched slated roofs, this methodology could potentially be adapted to other building fabric elements, such as walls.

3. Approach overview

As illustrated in Fig. 3, the proposed approach comprises three successive, fully automated stages:

A. Roof panel orthophoto generation;

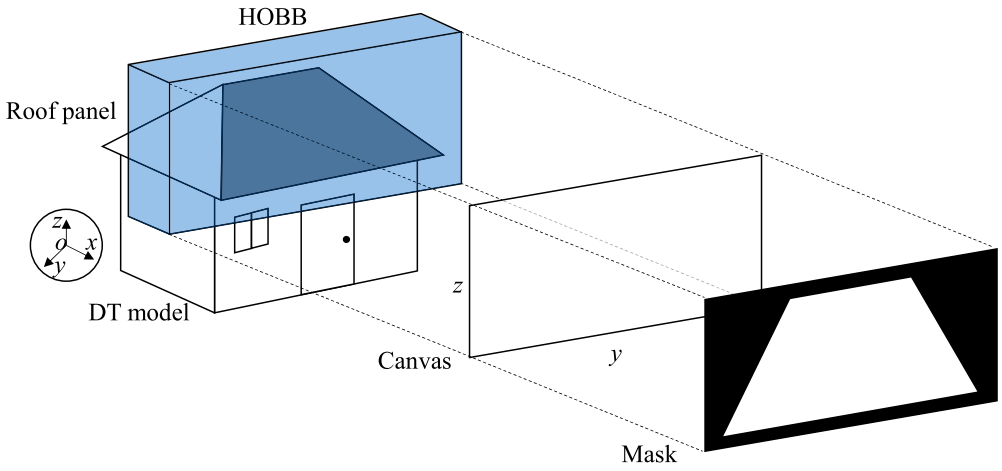


Fig. 4. Orthophoto generation example sketch map.

B. Roof panel orthophoto semantic segmentation; and
C. Deterioration and defect detection in the semantic segments.

The process takes two inputs:

- A structure-from-motion photogrammetric reconstruction of a roof that is referenced in the building's local coordinate system; and
- The Digital Twin 3D model of the building being monitored. The model must identify each individual roof panel as an element, with its own geometry and other semantic identification.

The output of stage A is an orthophoto for each roof panel, with minimised blur and without projection artefacts. Stage B then segments each one of these orthophotos semantically into its main material sub-components: slates, leadwork, and other. These segmented areas are then processed in Stage C using dedicated algorithms to output deterioration and defects related to that type of material. The detected deteriorations and defects are then systematically recorded (including with their exact position) in the building's Digital Twin. The scope of this paper is Stage A and Stage B. Stage C is not covered.

Stage A's method is detailed in Section 4 with associated experimental validation reported in Section 5. Stage B's method is then detailed in Section 6 with associated experimental validation reported in Section 7.

4. Roof panel orthophoto generation

The roof orthophoto generation method is composed of 3 main successive steps: 1. Creating panel orthophoto templates, 2. Selecting the most suitable images, 3. Instantiating orthophoto. Step 1 is conducted once offline, while Steps 2 and 3 are conducted each time an image acquisition campaign is conducted by UAV survey for that roof, for example, once a year in a proactive maintenance regime. Step 2 selects from the complete set of UAV-taken images, those images most suitable to generate an orthophoto for each roof panel. Step 3 then processes those selected images to 'texture' each panel orthophoto template, thereby instantiating an orthophoto from the given data acquisition campaign. These three steps are detailed in Sections 4.1 to 4.3, respectively.

4.1. Creating panel orthophoto templates

The first (offline) step for generating orthophotos is to build an *orthophoto template* for each roof panel. As illustrated in Fig. 4, each panel orthophoto template is defined by a *canvas* with associated *mask*:

- The *canvas* is the vertical face of the Horizontally-Oriented Bounding Box (HOBB) that is parallel to the main horizontal direction of the roof. The HOBB is the Bounding Box of the object that is object-oriented in the horizontal plane and axis-oriented vertically. The pixel resolution of the orthophoto image canvas is set so that each pixel is square with dimensions $\delta \times \delta$, e.g. $\delta = 5$ mm.
- the *mask* of the orthophoto template is defined by projecting all the panel's mesh triangles onto the canvas, which defines the extent of the roof panel within the canvas. In the mask, the RGB value of each pixel outside the projected triangles is set to [0, 0, 0] (black), and those inside the projected triangles is set to [1, 1, 1] (white).

This process is illustrated in Fig. 4 for one roof panel.

4.2. Selecting most suitable images

Before filling the masked area of each panel template created in Section 4.1, it is essential to select suitable UAV-taken images. This necessity arises because the original images were captured from varying incidence angles and locations, which can lead to resolution artefacts e.g., the target panel appearing too small, and the inclusion of pixels from occluding elements. To exclude images that are too far or oblique, which would only negatively contribute to the final orthophoto, they are filtered out with three parameters set below with an example illustrated in Fig. 5.

We define the ray shot from the pinhole camera O_i through the image centre C_i, and the intersection point with the roof plane is denoted as C_i'. Let us call C_p the centre of the target panel, and $\vec{n_p}$ its normal. Given this, the filtering parameters considered in this study are:

- α_{pi}: the angle between $\overrightarrow{C_i'O_i}$ and $\vec{n_p}$, which captures the picture incidence angle between respect to the roof panel.
- d^1_{pi}: the distance $\|\overrightarrow{O_iC_i'}\|$, which broadly captures the distance from the camera to the roof panel along the view direction.
- d^2_{pi}: the distance $\|\overrightarrow{C_pC_i'}\|$ normalised by the maximum dimension of this panel d_p^{max}, which can be used to measure whether the camera is pointing at the target panel or away from it.

The smaller α_{pi}, d^1_{pi} and d^2_{pi} are, the better the capture of the roof panel in the image should be. The thresholds for the three parameters are set as follows:

- α_{pi}: the threshold is arbitrarily set as 45°, beyond which images are defined as 'too oblique'.
- d^1_{pi}: the threshold is set as twice the value of the average UAV flying height from the roof.
- d^2_{pi}: the threshold is set as 1, which is exactly the value of the largest dimension of the roof.

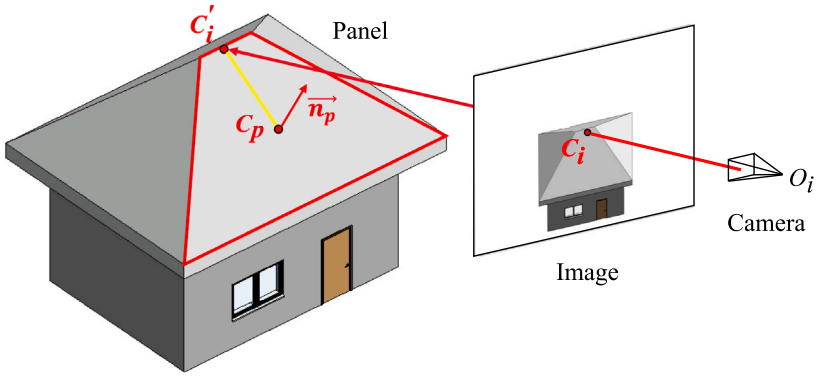


Fig. 5. The different parameters for screening input images for the generation of roof panel orthophotos.

4.3. Instantiating orthophoto

Each selected image is then projected onto the templates, to texture and instantiate individual orthophotos. Then, for each panel, a final orthophoto is generated by merging the individual orthophotos.

4.3.1. Individual orthophoto

The panel content inside the UAV image is first framed out by its boundary, to reduce the processing time. To realise this, each panel boundary of the DT model in the world coordinate system is transformed into the camera's 3D coordinate system (by camera external calibration, using Eq. (1)), and then into the image coordinate system (by camera internal calibration, using Eqs. (2) and (3)):

1. **From world coordinate system to camera coordinate system.**

$$\mathbf{P_c} = \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathrm{t} \end{bmatrix} \begin{bmatrix} \mathbf{P}_w \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathrm{t} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \tag{1}$$

where $\mathbf{P_c}$ and $\mathbf{P_w}$ are the point coordinates in camera system and world system respectively. $\mathbf{R}$ represents the 3×3 rotation matrix in the world system, defined by the three camera rotation angles around x, y, and z axes. t represents the 3×1 camera translation vector. Possible polynomial distortion transformations in some camera system should be applied to normalise $\mathbf{P}_c$ by dividing its first two components, X_c and Y_c, by its last one Z_c.

2. **From camera coordinate system (3D) to image coordinate system (2D).**

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{\mathbf{Z_c}} \mathbf{K} \mathbf{P_c} \tag{2}$$

$$\begin{cases} \text{scale} = \max(\text{w}, \text{h}) \\ \text{x} = \text{scale} * u + w/2 \\ \text{y} = \text{scale} * v + h/2 \end{cases} \tag{3}$$

In Eq. (2), $\mathbf{K}$ is the 3×3 camera's internal calibration matrix, containing focal length, camera skew, aspect ratio, and principle point offset. u and v, in the range $[-0.5, 0.5]$, are relative coordinates in the image system (centre as the origin). In Eq. (3), w and h are the width and height pixels of the image. x and y compose the pixel coordinate in the image system (top left corner as the origin), which is the final output.

With this process, the boundary polygon is projected onto the UAV images, then pixels to be projected from the UAV image to the DT model are limited to those within the projected panel boundary. Then, the texturing process on the orthophoto plane follows 2 steps:

1. **2D-3D.** All the 2D pixels within the original image panel (within the projected panel boundary) are projected to the DT mesh data using the camera calibration matrices. This outputs corresponding 3D points.
2. **3D-2D.** All the 3D points derived are then projected onto the orthophoto template.

This process textures the masked area of each roof panel with pixels from each image from the selected image set from Section 4.2, thus outputting *individual* orthophotos, i.e. one orthophoto for each input UAV image.

4.3.2. Final orthophoto

While individual orthophotos can sometimes be used as-is for analysis, they may not encompass the entire roof or may be marred by occlusions. To remedy this, all individual images for a given panel are aligned and merged to produce a single comprehensive orthophoto. This process is achieved as follows:

1. **Aligning orthophotos by Scale-Invariant Feature transform (SIFT) feature matching.** To eliminate the small shift between orthophotos caused by the small errors from the photogrammetric process and avoid its influence on the merging process, SIFT features are detected and matched across images [67]. These matchings are then used to calculate homographies to shift all individual orthophotos to one baseline one (selected as the orthophoto from the most suitable UAV image as defined in Section 4.2). Fig. 6 illustrates this alignment process and its benefit on the final *merged* image.
2. **Merging all orthophotos for one roof panel into one.** To exclude pixels from occluders, the aligned individual orthophotos can be merged by calculating the median value of the RGB for each orthophoto pixel. If, as we have observed, occlusions appear in fewer than 50% of the images, merging the orthophotos helps eliminate occlusions.

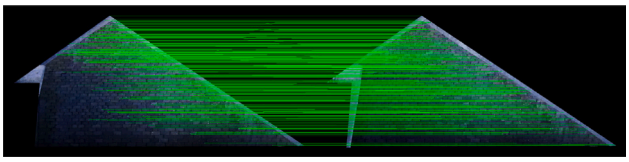
The above process aims to generate a final orthophoto of good clarity and occlusion-free.

5. Experimental results of orthophoto generation

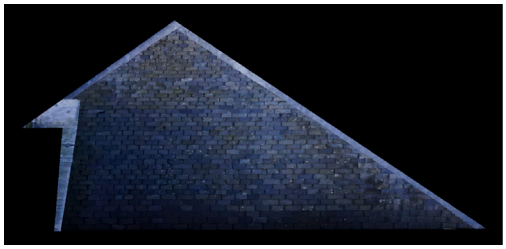
5.1. Experimental data

Two case study buildings are included in this study:

Duff House : Duff House is a historic building located in Banff, Aberdeenshire, Scotland. It is a category A listed, early Georgian mansion under the care of Historic Environment Scotland (HES) [68]. This building is a complex one containing different pitched roof panel forms (flat, curved, etc.), shapes (triangle,



(a) SIFT feature matching.

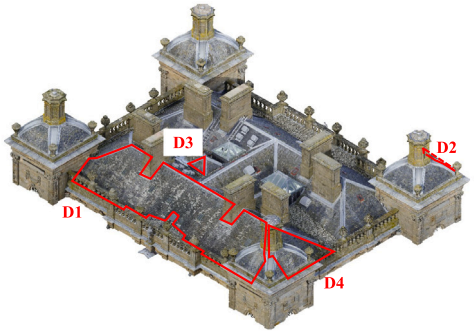


(b) Orthophoto before alignment.

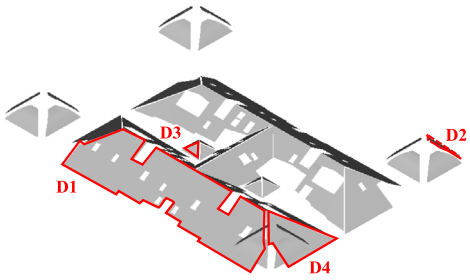


(c) Orthophoto after alignment.

Fig. 6. Pixel shifting.



(a) Structure-from-motion photogrammetric reconstruction (3D point cloud as well as calibrated images).



(b) Roof panel geometries in the building's Digital Twin.

Fig. 7. Duff House input data to our proposed process.

trapezoidal, parallelogram, etc.). It also contains other components such as chimneys and balustrades, the building dimension is 34.2 m in length and 25.7 m in width. Due to its data complexity, this example is suitable for training and shaping the generalisation of the segmentation model.

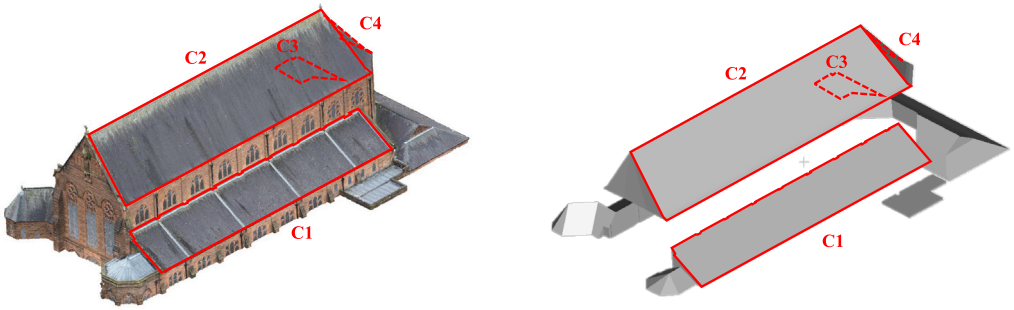
Duff House was fully surveyed using a UAV (DJI Mavic 2 Pro, same for St Mary's Church) equipped with a Sony ILCE-7R digital camera, with a fixed focal length of 35 mm. The UAV survey acquired 859 images containing the roof, and with over 75% overlapping, enabling a detailed and accurate 3D reconstruction of the roof (and upper parts of the façades). A full 3D reconstruction of the house exterior (façades, entrance stair case, etc.) was also obtained by merging the UAV reconstruction with a Terrestrial Laser Scanning (TLS) point cloud.

The structure-from-motion photogrammetric reconstruction of Duff House is shown in Fig. 7, alongside the 3D DT model of its roof. The resulting 3D point cloud contains 2,104,179 points and is registered in the world coordinate system. The point cloud density is approximately 2400 points/m^2. The roof DT model contains 36 individual slated panels (20 planar ones and 16 curved ones). The required input DT model is a more granular model than BIM model, appearing as an Industry Foundation Classes (IFC) object for each individual roof panel, finished using the photogrammetric reconstruction as a baseline. For historic buildings, such Scan-to-BIM process is a simulation of the original building designing task. IFC model contains geometry, location and other necessary information of each roof panel. The geometry of each panel is defined as panel boundary with extrusion. As input to the proposed process, geometries of each roof were then defined as triangle meshes.

St Mary's Church : St Mary's Church is a Roman Catholic parish church, located in Stirling, Scotland. It is a category B listed Highland church [69]. This building has a two-story pitched roof that contains large rectangular panels spaced apart by lead strip, which can help test the robustness of the trained model. The dimension of main part of the building is 44.7 m in length and 25 m in width. This building was surveyed using the same UAV as Duff House but equipped with another Hasselblad l1d-20c camera, with a fixed focal length of 28 mm. 1245 images were obtained for reconstructing the roof with 75% overlapping.

The structure-from-motion photogrammetric reconstruction of St Mary's Church is shown in Fig. 8: The 3D point cloud contains 1,570,136 points and is registered in the world coordinate system. The point cloud density is approximately 1255 points/m^2. This roof DT model contains 18 individual slated panels, all of them are planar.

The surveying work (including flying the drone and conducting the 3D reconstruction) for those case study buildings took in the order of several hours.



(a) Structure-from-motion photogrammetric reconstruction (3D point cloud as well as calibrated images).

(b) Roof panel geometries in the building's Digital Twin.

Fig. 8. St Mary's Church input data to our proposed process.

5.2. Evaluation metrics for orthophoto generation

The orthophoto generation results are evaluated qualitatively and quantitatively, considering:

- **occlusion**: the extent to which occlusions from other components have been removed (qualitative);
- **blur**: the general level of blur the orthophoto presents, which is evaluated by assessing the sharpness of the boundary of each slate (qualitative);
- **merging quality**: the quality of the final merged orthophoto for each panel, measured by the mean percentage of matched features, $\bar{a}$ calculated using Eq. (4) (quantitative).

$$\bar{a} = \frac{1}{n-1} \sum_{i \in \mathcal{O} \setminus \{b\}} \frac{|\mathcal{F}_i^b|}{|\mathcal{F}_b|} \tag{4}$$

where $\mathcal{O}$ is the set of n orthophotos associated to the given panel, b is the baseline orthophoto, and $\mathcal{F}_i^b$ is the set of SIFT features from orthophoto i matched to the baseline orthophoto, and $\mathcal{F}_b$ is the set of SIFT features from orthophoto b.

5.3. Results

5.3.1. Value of image selection and fine alignment on merged orthophoto quality

The stages in the generation of an orthophoto, including the final refinement, are demonstrated in Fig. 9, using roof panel D4 as an example: Fig. 9(a) shows four typical individual orthophotos generated from images potentially containing some information about panel D4: the upper left one is the best of the four, but the lighting condition varies across the panel; the upper right one is of lower quality, mainly due to the presence of an occluder (the chimney in this example) and artefacts; the lower two individual orthophotos are from pictures that only partly capture the panel, and one of them even contains pixels from an occluder (a nearby tower).

The above images are 4 of 78 images that seemingly contain information about panel D4 before the image selection process is applied. Fig. 9(b) shows the result of merging the 78 individual orthophotos obtained from those 78 UAV images. As can be seen, this resulting orthophoto is free of pixels from occluders and also does not contain any hole (i.e. empty pixel) in the roof panel area, which shows the value of the median value merging strategy. However, careful observation shows that this merged orthophoto is blurry.

When the image selection process explained in Section 4.2 is applied, 67 of the 78 images are discarded, and only 11 are retained. Such selection is highlighted in both 3D and 2D scatter plots in Fig. 10, in which the red dots represent the selected images, while the black dots are the remaining ones. The merged orthophoto obtained from the selected 11 images is shown in Fig. 9(c). It contains less but still some blur than that of the initial merged orthophoto in Fig. 9(b).

Finally, Fig. 9(d) shows the median merged orthophoto obtained using only the selected set of images and after fine alignment. It has significantly less blur than the median orthophoto in Fig. 9(c) obtained after merging and image selecting but without fine alignment.

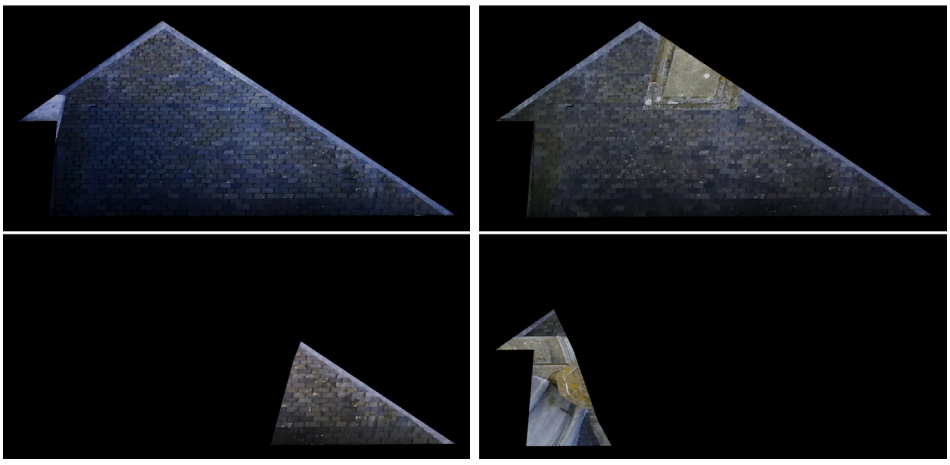
5.3.2. Quality of final orthophotos

Four representative panels in Fig. 7 are used for reviewing the application of the proposed orthophoto generation method: Panel D1 is the largest panel within the whole roof, placed on the front and back side of the house (easily occluded by balustrades, chimneys and towers in some images, it does not appear completely in any single image). Panel D2 is a curved panel placed on the one of the towers of the house (easily occluded by stone ornaments). Panel D3 is one of the triangular panels placed near the roof centre with almost no occlusion. Finally, Panel D4 is one of the panels placed on the side of the house (easily occluded by the towers and balustrades on the edges). The results obtained for all four panels are shown in Fig. 11.

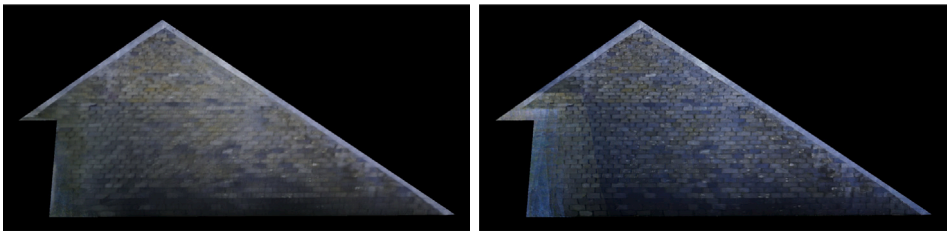
Panel D1 presents the worst result, with the most significant and uneven level of blur. For this panel, $\bar{a} = 10\%$, which is the lowest among all the panels. This is likely due to the large size of the panel, which results in the whole panel not being seen completely in any single selected image. As a result, the alignment process described in Section 4.3.2 starts with one image that only covers a small part of the roof. Notwithstanding, in comparison to Fig. 2 where the chimney occludes part of the roof, all the potential occlusions for this panel are successfully removed. The result for the curved Panel D2 is better, with $\bar{a} = 35\%$. This is likely because it is smaller and not as occluded, so that the baseline orthophoto has more matches with the other orthophotos. In the smallest Panel D3, the details are more clear under magnification: defects like the shallow organic growth appear very clearly. In this case again, $\bar{a} = 38\%$. For Panel D4, $\bar{a} = 25\%$, which is also higher and also correlates with a final orthophoto of higher quality.

These orthophoto examples demonstrate the generally good performance of the proposed approach. Indeed, at least for panels of small to medium sizes (e.g. up to 30 m^2), the slated roofs are clearly and precisely extracted and should thus be well suited for semantic segmentation (as explored in Section 6) and defect detection. Besides, no matter which angle was the image taken or where the panel is located, the final orthophotos are all in the uniform scale and vertical, with the slates appearing as parallel horizontal rows in all images. This should further ease the development and application of robust automated defect detection models — and such representation also remains suited for manual defect annotation using current manual methodologies.

The method however performs less well for larger panels. To tackle this, they could simply be split into sub-areas within which the proposed method will work. Alternatively, the orthophoto image merging

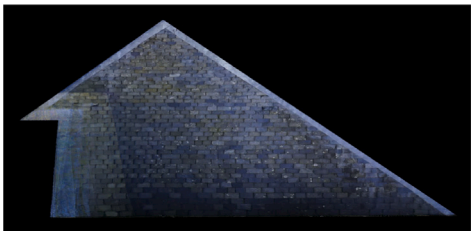


(a) Individual orthophotos for roof panel D4: one with imbalanced lighting (upper left), one with occlusion and artefacts (upper right), two only partly captured (lower left and right)



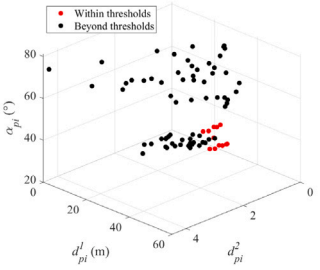
(b) Initial median-merged orthophoto for roof panel D4.

(c) Median-merged orthophoto for roof panel D4 after application of the selecting process.

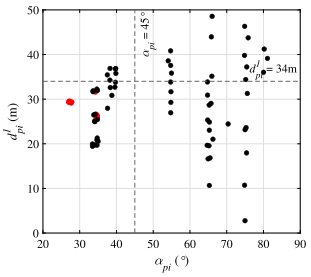


(d) Median-merged orthophoto for roof panel D4 after application of the selecting and fine alignment processes.

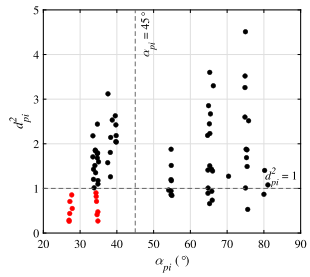
Fig. 9. Orthophoto transformation process for roof panel D4 from individual orthophotos to merged ones using different datasets.



(a) α_{pi}, d^1_{pi} and d^2_{pi} in 3D scatter plot



(b) α_{pi} and d^1_{pi} in 2D scatter plot

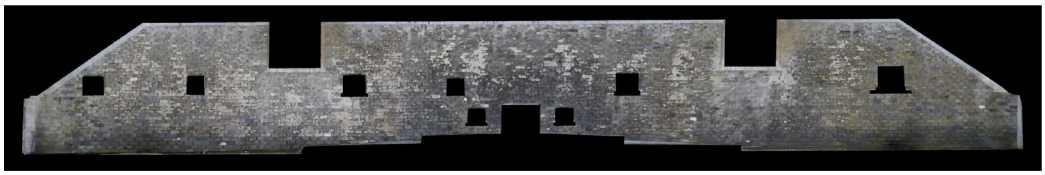


(c) α_{pi} and d^2_{pi} in 2D scatter plot

Fig. 10. α_{pi}, d^1_{pi} and d^2_{pi} in different images in 3D and 2D scatter plot form.

strategy should be reviewed and developed so that the refinement is conducted more globally than from the localised data associated with the baseline photo.

Another four representative panels from the St Mary's Church (Fig. 8) are illustrated in Fig. 12. The $\bar{a}$ values for the four panels are: 6%, 8%, 7% and 10%, and their widths are 25.4 m, 27.1 m, 6.9 m and 3.1 m. The longer panels, Panel C1 and Panel C2, have relatively lower $\bar{a}$ values, originating from the blur problem occurring on the left and right sides of these panels. Panel C3 is a relatively smaller panel, so there is less blur. However, by comparing the filtering parameter scatter plots of this panel in Fig. 13 to the ones in Fig. 10, it can be easily seen that the distributions of parameters, especially the d^1_{pi}



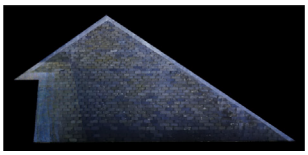
(a) Panel D1.



(b) Panel D2.



(c) Panel D3.

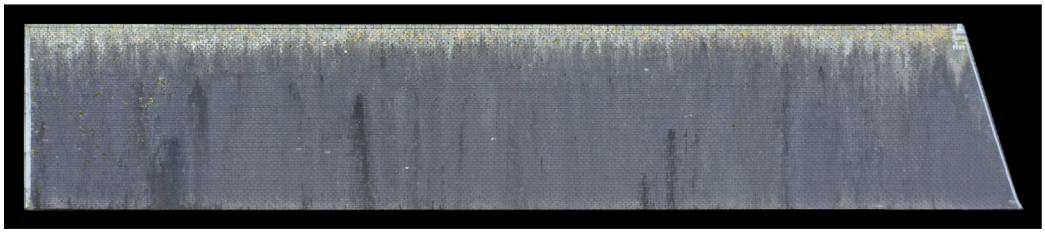


(d) Panel D4.

Fig. 11. Generated orthophotos of typical panels.



(a) Panel C1.



(b) Panel C2.



(c) Panel C3.



(d) Panel C4.

Fig. 12. Generated orthophotos of typical panels of St Mary's Church.

and d_{pi}^2 are broader, which implies that some of these images have low overlapping with the selected baseline orthophoto in the merging process, ultimately making the $\bar{a}$ value low. Notably, Panel C3 was frequently occluded; yet the occlusions are successfully removed in the final orthophoto. Finally, Panel C4 has is much smaller, leading to even better results. Correspondingly, its $\bar{a}$ value is higher.

6. Semantic segmentation

The input to this stage is an orthophoto of a slated roof panel generated by the method presented in Section 4. The generated orthophoto shows the roof panel that is composed of different essential sub-components: slates and leadwork, as well as other less frequent elements (e.g. stone, grid, equipment, glass, ladder, etc.).

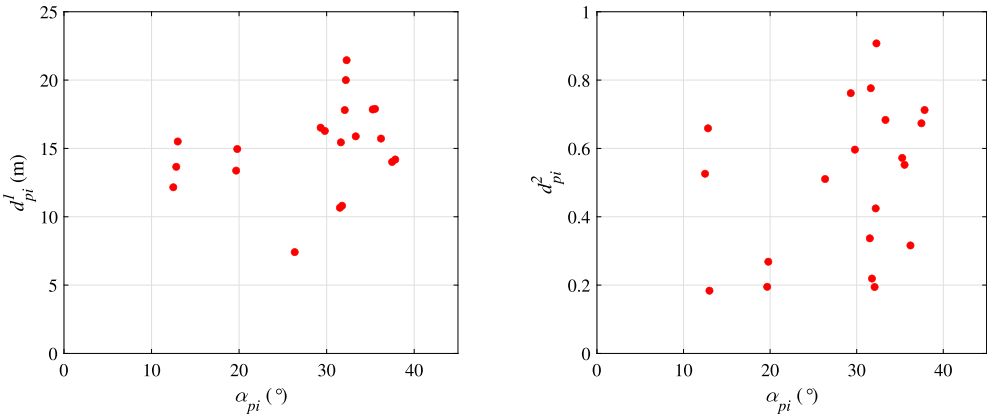
Since the intention of this work is to find defects in slated areas as well as leadwork areas, and these materials have distinct types of defect types, it is best to first segment the orthophoto to isolate these particular subparts. For this, we explore different methods for (pixel-level) semantic segmentation.

6.1. Dataset

The data output of Duff House by Li et al. [14] includes data for 36 panels. We split these into 27 panels for training and 9 panels for testing. For each panel, we retain maximum 10 unmerged orthophotos obtained from different photos of the same panel (with different angles and covering the panel in various ways). This dataset is used for developing and comparing different initial semantic segmentation models.

In addition, the St Mary's Church data include data for 18 panels. With this data, we create another dataset composed of one unmerged orthophoto per panel. This dataset is used solely to test the generalisability of the models created using the Duff House dataset.

Finally, a combined dataset using data from the two buildings is created. It is divided into a training dataset that is ~80% of the overall dataset (27 panel orthophoto data of Duff House, along with 13 panel orthophoto data of St Mary's Church) and a testing dataset that is the remaining ~20% (9 from Duff House, along with 5 from St Mary's Church). Here, only one unmerged orthophoto is kept for each panel. We select the orthophoto with the largest coverage of the panel.



(a) α_{pi} and d^1_{pi}. (b) α_{pi} and d^2_{pi}.

Fig. 13. Filtering parameters distributions of Panel C3.

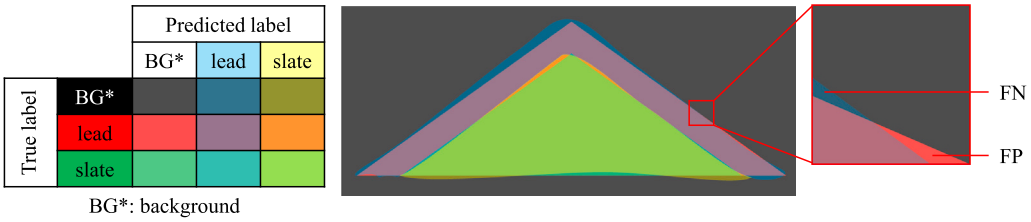


Fig. 14. Colour coded confusion matrix (left), semantic segmentation model output (middle), illustrations of FP/FN from the perspective of *background* class (right).

Table 1
Experimental parameter settings for the deep learning model training.

Parameter	Value
Batch size	4
Max iteration	2400
Validation interval	400
Training vs. validation	80% : 20%

All the orthophotos used in the datasets above are manually labelled in 4 classes: *background* (labelled as 0), *slate* (1), *leadwork* (2), and *other* (3). The *other* category contains: stone, grid, equipment, glass, ladder, etc. The class *other* normally occupies a very small proportion of pixels in orthophotos.

6.2. Deep learning model

As discussed earlier in Section 2, state-of-art models that can be useful to the specific problem in this study include: *DeepLabV3*, *PointRend*, *Segmenter*, *Segformer*. Basic hyper-parameter settings are tuned for best performance by comparing these deep learning models. The best settings are reported in Table 1. All models were pre-trained using their default weights.

For each model, K-fold cross validation is applied during training: the entire training dataset is first split into k folds ($k = 5$ in this study). The model is trained k times on each $k-1$ folds, and then validated on the remaining fold. The model performances are evaluated using the average value of these k time validation results, and the best model is selected based on those. Then, the best model is retrained using the entire training dataset, and the final result is obtained by testing this model on the testing dataset.

During training, the input images are not rescaled. This is because most of the roof orthophotos are rectangular with varying width-to-length ratios, and the resizing processing operations that are typically applied in semantic segmentation pipelines would result in information loss. But, to meet the requirements of input image size and computing capacity limits, all images are cropped (tiled) and read as $512 \times 1024 \times 3$ matrices.

As will be shown in the Experimental Validation using the Duff House dataset (Section 7.3), PointRend with default backbone (ResNet) and loss function (Cross-Entropy Loss) achieves best performance all four initial models. As a result, further experiments are conducted by using different loss functions including Dice Loss (sensitive to region detection) and Focal Loss (sensitive to imbalance problem). With grid search as a tool for hyperparameter adjustment, specific weights between different losses are selected for best performance.

The models trained with the Duff House dataset were then tested with the St Mary's Church dataset to assess its generalisation ability. The results lead us to finally use the combined orthophoto dataset (Duff House + St Mary's Church) for training and testing a final model with best segmentation performance and generalisation ability.

6.3. Background label correction

In most deep learning semantic segmentation methods, precise delineation of segmentation boundary is a challenging problem, with many confusions arising at those boundaries. Fig. 14 illustrates this issue with a typical output of the semantic segmentation models we explored. The figure highlights the discrepancies between the *Prediction* and *Ground Truth* (*GT*) for the different classes.

However, in this study, the set of *background* pixels is actually known a priori, because the panel boundary is defined by the Digital Twin 3D model projection (see [14] for details). In other words, the Ground Truth for the *background* class is known a priori. Therefore, an extra step is introduced to correct the False Positive (FP) and False Negative (FN) results for the *background* class:

1. False Negative ($Prediction \neq 0$ and $GT = 0$): In this case, the predicted label is simply changed back to '0' (i.e. *background* class).

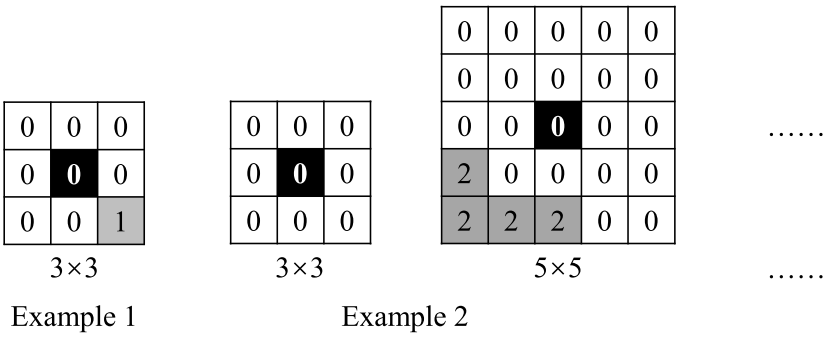


Fig. 15. Illustration of the correction of FP for the *background* class. Left: Example 1 where the label is corrected to '1' after one step; Right: Example 2 where the label is corrected to '2' after two steps.

2. False Positive ($GT \neq 0$ and $Prediction = 0$): In this case, the predicted label is changed from the *background* class to the most likely *other* class. As illustrated in Fig. 15, for each FP pixel the non-zero label that appears most frequently in the 3×3 grid around it is selected as the new label. If the grid contains only *background* pixels (i.e. class '0'), then the grid is expanded by one pixel (i.e. 5×5 grid) and the process is repeated until a at least a non-zero label is found. As will be shown, this simple process works well in our context.

As will be shown in the Experimental Results (Section 7), while the baseline semantic segmentation results are already good, this process delivers some additional improvements.

6.4. Label merging strategy

As we have seen in Section 4, each individual orthophoto may not necessarily cover entirely a given roof panel of interest. Besides, the textured parts of multiple individual orthophotos will most likely overlap. Ultimately, we do not wish to have many orthophotos (with their individual semantic segementation) for each roof panel, but a single one covering the overall panel with a single semantic segmentation. This can be achieved using either of the following two merging strategies:

- *Label Merging Strategy 1*: Merge the unmerged orthophotos as described in Section 4, and then apply the semantic segmentation to the merged orthophoto.
- *Label Merging Strategy 2*: Merge the semantic segmentation results of all unmerged orthophotos. For each pixel, the label that appears most frequently (top 1 class) among the unmerged orthophotos is selected as the final label. In cases when two (or more) classes have the same frequencies for all unmerged orthophotos, the final label will be selected in the following order of priority: *other*, *leadwork*, *slate*. For example, if *slate* and *other* appear the same time for one pixel, then *other* would be chosen as the final label.

These two merging strategies are compared on the model trained with the combined dataset.

7. Experimental results

In this section, evaluation metrics for semantic segmentation are first introduced. Then all the experimental results are reported. Different models are trained using the Duff House training dataset and tested using the Duff House and St Mary's Church testing datasets. The best model is selected by comparing these results, and it is finally re-trained and tested using the combined dataset. All the training and testing work is completed in the Google Colab Pro environment, with NVIDIA A100 GPU 40 GB.

7.1. Evaluation metrics for semantic segmentation

The segmentation results for each class are evaluated using two parameters:

- *Intersection over Union (IoU)*: computed to assess the overlap between Prediction and Ground Truth segments.
- *Accuracy (Acc)*: calculated by dividing the sum of the True Positive pixels by the sum of the True Positive pixels and False Positive pixels.

To compare the general testing performance between different models, the following overall evaluation metrics are evaluated:

- *aAcc*: the Accuracy of all pixels, evaluating the classification accuracy.
- *mIoU*: the mean *IoU* of all classes; *mIoU* is an important indicator to measure the accuracy of overall semantic segmentation.
- *mAcc*: the mean *Acc* of all classes, evaluating the overall performance in pixel classification.

In the following, we report results for different models in the form of tables and confusion matrices. In the tables, we report for each model: the overall *aAcc*, *mIoU* and *mAcc*, and then *IoU* and *Acc* for each of the four classes. We also report relative (in percentages) confusion matrices.

7.2. Background label correction

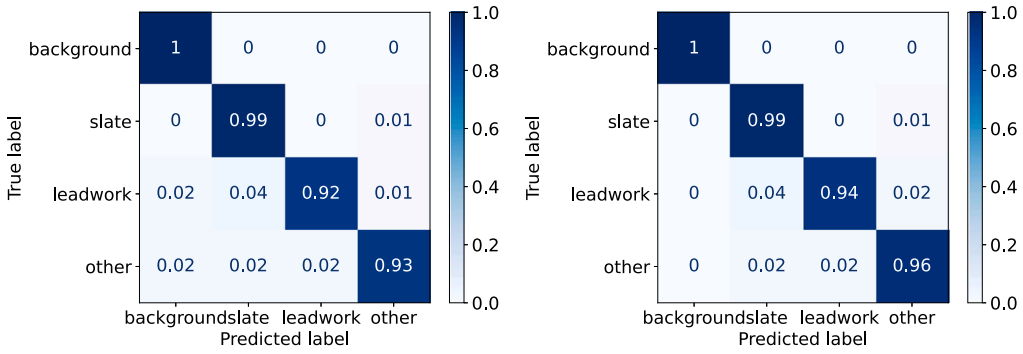
Regardless of the model employed, the *Background model correction* step described in Section 6.3 can be applied to correct FP and FN for the *background* class. We thus demonstrate the benefits of this correction using one model (which we will see later performs well): PointRend.

Table 2 and Fig. 16 show the results obtained when training and testing this model on Duff House dataset, before and after applying the *background* label correction results.

While the baseline performance is already quite good (all metrics >85 and most of them are >95), the additional corrective step improves performance for all classes, in particular the *leadwork* and *other* classes, which had the lowest performance without this correction. The higher increases in mIoU (+1.58 to 94.22) and mAcc (+1.15 to 97.25) also indicate a reduced difference in performance among the different classes. Although anticipated, this improvement is welcome, because the *background* regions are often next to *leadwork* regions. And so, any correction of *background* class would most likely benefit the *leadwork* class. Nonetheless, the results demonstrate the good performance of the proposed *background* label correction method.

Table 2
Testing results before and after background label correction.

Result	Overall			Overall (excl. *background*)			*background*		*slate*		*leadwork*		*other*	
	aAcc	mIoU	mAcc	aAcc	mIoU	mAcc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Before	98.93	92.64	96.1	–	–	–	99.26	99.79	98.09	98.87	88.49	92.27	84.7	93.5
After	99.31	94.22	97.25	98.40	92.29	96.34	100	100	98.44	99.17	91.24	94.31	87.19	95.53



(a) *Before.* (b) *After.*

Fig. 16. Confusion matrices before (a) and after (b) applying the *background* label correction step.

Table 3
Comparing trained models (with K-fold cross validation) on Duff House.

Model	Overall			*background*		*slate*		*leadwork*		*other*	
	aAcc	mIoU	mAcc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
DeepLabV3	98.74	93.01	96.12	98.86	99.62	97.44	98.41	86.20	91.85	89.53	94.60
PointRend	**99.13**	**94.77**	**97.27**	99.31	99.79	**98.13**	98.82	**90.29**	94.14	**91.35**	**96.36**
Segformer	98.79	93.17	96.71	99.05	99.64	97.19	98.18	87.48	93.18	88.96	95.83
Segmenter	98.20	90.70	94.27	98.36	99.46	96.20	97.84	80.80	88.26	87.44	91.53
PointRend(CEL+DL)	99.08	93.82	96.30	**99.40**	99.77	98.03	**99.04**	89.49	**94.22**	88.35	92.15
PointRend(CEL+FL)	99.03	94.01	96.62	99.31	**99.80**	97.90	98.83	89.53	93.33	89.28	94.53
PointRend(DL+FL)	99.06	94.14	96.59	99.29	99.76	98.06	99.00	89.06	93.02	90.14	94.54

7.3. Initial models with Duff House dataset

All models are first trained using Duff House training dataset only, and using K-fold cross validation. Table 3 presents the validation performance of the different models. Generally, all models already show good performance: the evaluation metrics of overall performance are all >90, most of them are >95 and even nearly 100. Generally, the target class, *slate*, is segmented satisfactorily. Errors mostly come from the classes *leadwork* and *other*.

By comparing the first 4 rows, PointRend stands out with the highest values in all evaluation metrics. The variants in the last three rows are then developed based on PointRend, in an attempt to enhance performance with regard to specific challenges with our dataset, namely data class imbalance (the *leadwork* and *other* classes occupy much fewer pixels than *background* and *slate*) and region ambiguity.

Among all the variant models, PointRend(CEL+DL) may be considered to achieve the best overall performance, as it slightly increases the values of Acc both on *slate* class (+0.22 to 99.04) and *leadwork* class (+0.08 to 94.22). But, its performance is still close to that of the default PointRend model and has some sacrificial drops on most metrics. Therefore, with the current Duff House training dataset, the default PointRend model may still be used as the best model for retraining and testing.

With the best model, the segmentation prediction results using models trained by different datasets (unmerged or merged Duff House orthophotos) are compared in Table 4: The results show that the model trained with the dataset of unmerged orthophotos and tested on the dataset of unmerged orthophotos has the best performance. And when testing on the dataset of merged orthophotos, then the model trained on the dataset of unmerged orthophotos still outperforms the one trained on the dataset of merged orthophotos. These results help determine that it is in all cases best to train the model on the dataset of unmerged orthophotos.

7.4. Testing generalisation with St Mary's Church testing dataset

Table 5 reports the evaluation results of best model, PointRend, on the testing dataset of St Mary's Church. This enables an assessment of the model's generalisation, since no data from St Mary's Church was used to train those models. Table 5 shows a similar pattern as Table 3: It can segment the *slate* area more accurately than *leadwork* and *other* classes, while the *other* class has the worst performance among all classes. However, compared to the performance on Duff House, there are decreases in the overall performance, such as a nearly 30 percentage point drop in mIoU and nearly 20 percentage point drop in mAcc. Compared to the corresponding performance values in Table 4, although the performance values for *slate* remain at the same level, those for *leadwork* and *other* are much lower.

Therefore, even though aAcc remains close to 100% and the general performance is acceptable, the generalisation ability of the models is limited. Put another way, unsurprisingly the model, when trained using data from only one building (Duff House), work but not sufficiently well for other buildings. Therefore, more data, especially with diverse features of *leadwork* and *other* classes, should be used for training.

7.5. Models trained with combined dataset

Giving to the overall still limited amount of data available, all the models above are all retrained using the combined training dataset using K-fold cross validation. The comparison of the different models

Table 4
Testing results of Duff House merged orthophotos.

Strategy	Overall			Overall (excl. *background*)			*background*		*slate*		*leadwork*		*other*	
	aAcc	mIoU	mAcc	aAcc	mIoU	mAcc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Train unmerged/test unmerged	**99.31**	**94.22**	**97.25**	**98.40**	**92.29**	**96.34**	100	100	**98.44**	99.17	**91.24**	**94.31**	**87.19**	**95.53**
Train unmerged/test merged	98.75	86.98	93.56	97.46	82.64	91.42	100	100	97.81	98.76	89.21	92.36	60.89	83.13
Train merged/test merged	98.53	84.06	88.5	97.02	78.75	84.67	100	100	97.44	**99.25**	86.26	90.92	52.55	63.83

Table 5
Testing results on St.Mary's Church by models trained using Duff House training dataset.

Model	Overall			Overall (excl. *background*)			*background*		*slate*		*leadwork*		*other*	
	aAcc	mIoU	mAcc	aAcc	mIoU	mAcc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
PointRend	94.65	63.31	78.54	91.61	51.08	71.39	100	100	92.99	93.96	36.32	61.36	23.94	58.84

Table 6
Comparing trained models (with K-fold cross validation) on combined dataset.

Model	Overall			*background*		*slate*		*leadwork*		*other*	
	aAcc	mIoU	mAcc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
DeepLabV3	97.95	80.92	86.78	98.32	99.58	97.19	98.52	77.07	86.92	51.10	62.09
PointRend	98.17	82.55	88.51	98.80	99.72	97.46	98.65	77.73	87.88	56.21	67.81
Segformer	98.25	**83.72**	**89.31**	98.61	99.51	97.27	98.64	**83.02**	**90.81**	55.99	**68.28**
Segmenter	97.58	79.11	83.90	97.91	99.35	96.14	98.76	74.58	83.94	47.83	53.57
PointRend(CEL+DL)	97.25	78.12	86.73	**98.97**	**99.73**	95.94	97.32	66.64	86.02	50.94	63.85
PointRend(CEL+FL)	**98.33**	83.21	88.46	98.82	**99.73**	**97.68**	**98.96**	79.61	88.19	**56.73**	66.95
PointRend(DL+FL)	97.72	80.62	86.56	98.89	**99.73**	95.84	97.88	76.38	85.49	51.39	63.14

Table 7
Testing result on combined testing dataset.

Model	Overall			Overall (excl. *background*)			*background*		*slate*		*leadwork*		*other*	
	aAcc	mIoU	mAcc	aAcc	mIoU	mAcc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
PointRend(CEL+FL)	98.82	86.97	89.7	97.66	82.62	86.27	100	100	97.98	99.62	86.94	93.58	62.95	65.6

in Table 6 shows that both Segformer and PointRend(CEL+FL) have outstanding performance. However, since our main goal is the segmentation of the slated area, the PointRend(CEL+FL) model is selected as the best model. It is retrained and tested using the combined testing dataset, and the final performance results after background label correction are shown in Table 7.

The model performances reported in Table 6 are almost at the same level as those in Table 3, whose performances were already very high. However, obvious reductions in performance are observed for the *leadwork* and *other* classes for all models. But, importantly, compared to the prediction results in Table 5, most metrics show significant improvements in Table 7. This implies that models trained with the combined dataset achieved a greater level of generalisability (they perform well on testing data from both Duff House and St Mary's Church) without significant drop in overall performance. Naturally, this does not mean the new models will work in all cases of slated roofs; much more diverse data would probably need to be collected for that. But, given the performance of the previous models on the St Mary's Church dataset, the new models are likely to already perform satisfactorily.

7.6. Label merging strategy

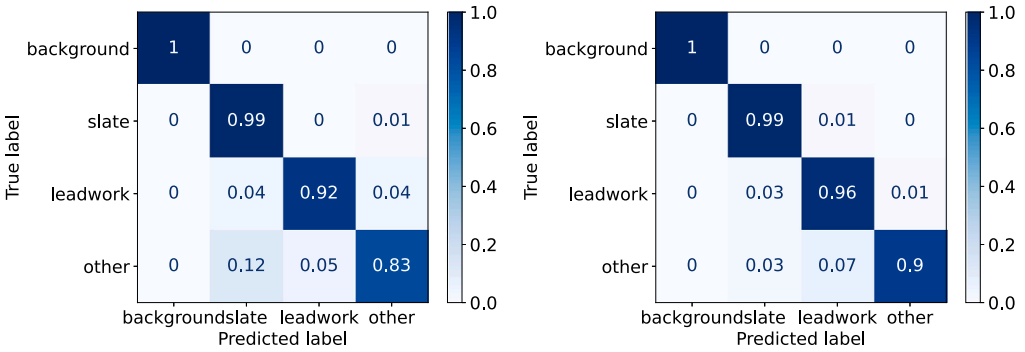
The model trained in the previous section is applied to the individual orthophotos generated from each image used in the 3D reconstruction. The results of the two label merging strategies presented in Section 6.4 are reported in Fig. 17.

Compared to the result of merging the unmerged orthophotos and then applying the semantic segmentation to the merged orthophoto (Strategy 1), merging the semantic segmentation results of all unmerged orthophotos (Strategy 2) shows superior segmentation performance: Accuracy is higher for *leadwork* (+0.04 to 0.96) and *other* (+0.07 to 0.9). More importantly, the confusion between *slate* and *other* classes is greatly reduced: in situation that True label is '*other*' but predicted as '*slate*', the error is reduced from 0.12 to 0.03, in situation that True label '*slate*' but predicted as '*other*', the error is also reduced from 0.01 to 0. Therefore, the better solution for obtaining the unified segmentation is to merge the semantic segmentation results obtained on individual orthophotos.

As can be seen from Fig. 18, most pixels, especially those of the *slate* class are classified correctly. The segmentation result of Panel D5 is nearly flawless. However, when the situation becomes more complex, for example the Panel D6 which includes some unavoidable projection of stone works on the orthophoto, then results worsen, with confusions between *leadwork* and *other* on the object edge, and some *slate* pixels are misclassified as *other*. This noise clustering is possibly caused by the overfitting problem by insufficient data. For Panel C1, there are pixels wrongly classified at the top of the roof panel, the *GT* of which is *slate* but they are predicted as *other*. The reason for such mistake possibly lies in the biological growth on the slate surface, which may confuse the deep learning model despite some of the training data containing it. This issue may nonetheless be addressed through a more extensive training dataset containing more examples of such situation.

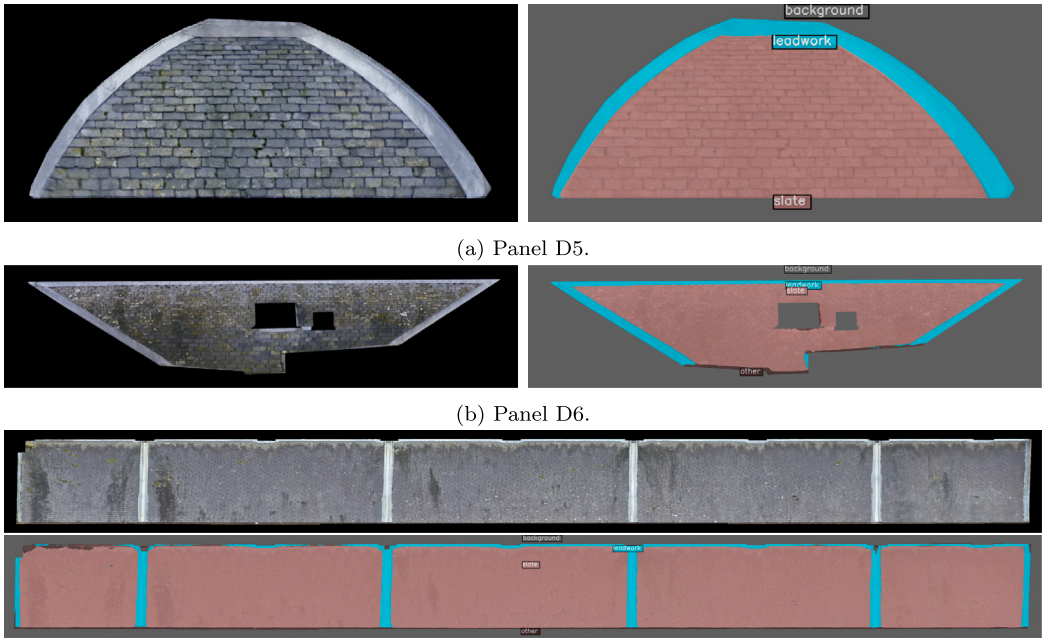
8. Validation using a case study: a residential house in Stirling

The complete pipeline proposed in this study is validated using another case study: a residential house located in Stirling, Scotland. This building was surveyed using DJI Mavic 2 Pro equipped with a Hasselblad l1d-20c camera, with a fixed focal length of 10 mm. 660 UAV images were acquired containing the roof, with over 75% overlapping. The point cloud data from photogrammetric reconstruction contains 588,105 points in world coordinate system, with the dimension of 17.4 m in length and 13.9 m in width, making the estimated point density 2431 points/m^2. The corresponding DT model of the building



(a) Strategy 1. (b) Strategy 2.

Fig. 17. Confusion matrices of strategy 1 (a) and strategy 2 (b).

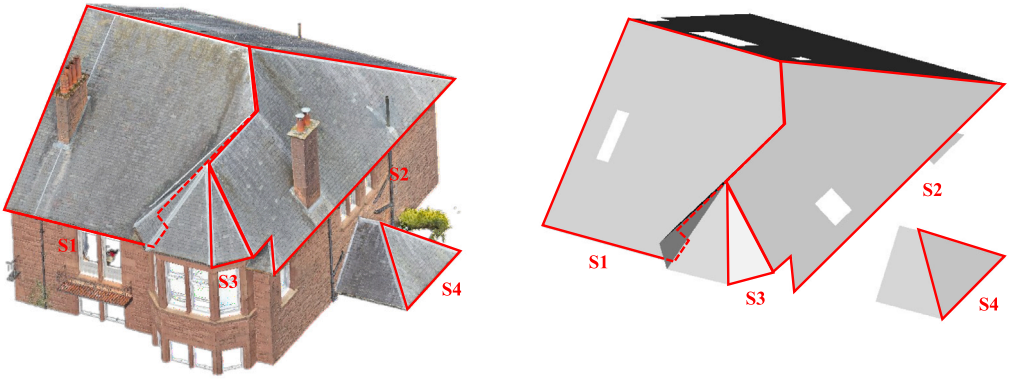


(a) Panel D5.

(b) Panel D6.

(c) Panel C1.

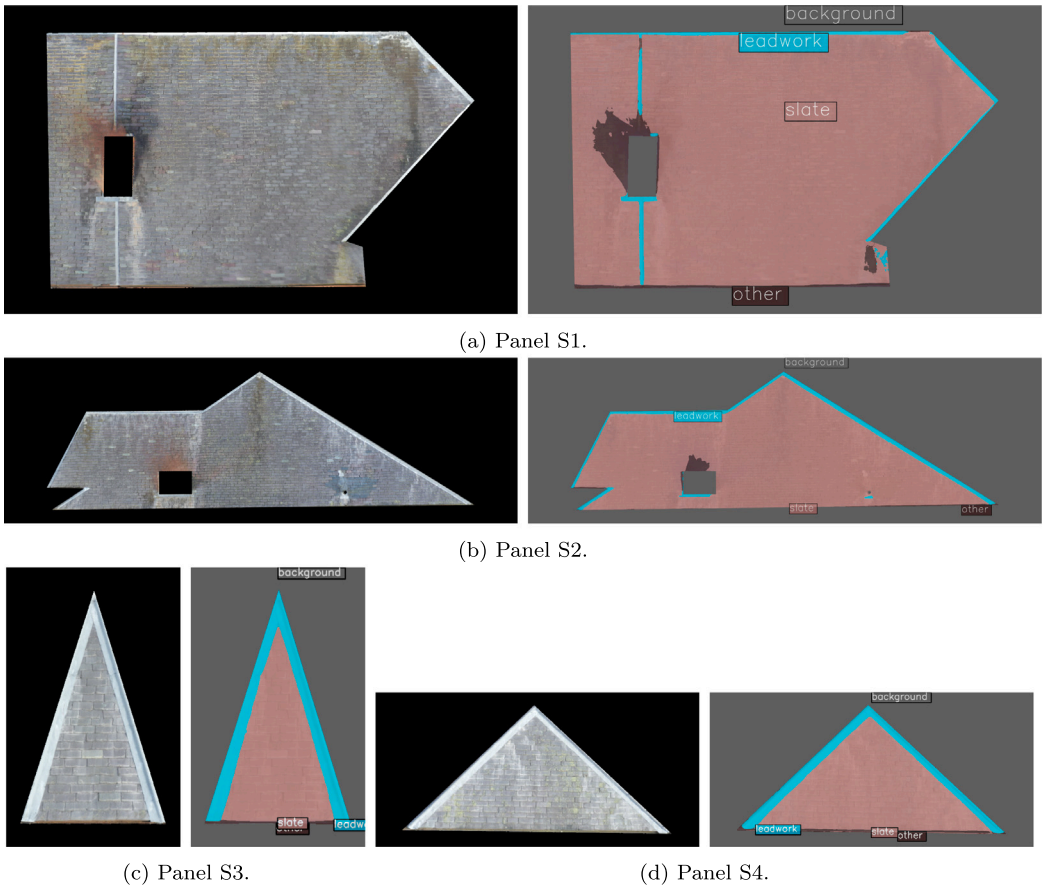
Fig. 18. Orthophotos and corresponding semantic segmentation results of typical panels.



(a) Structure-from-motion photogrammetric reconstruction (3D point cloud).

(b) Roof panel geometries in the building's Digital Twin.

Fig. 19. Residential house data input to our proposed process.



(a) Panel S1.

(b) Panel S2.

(c) Panel S3.

(d) Panel S4.

Fig. 20. Generated orthophotos and corresponding semantic segmentation results of typical panels of residential house in Stirling.

Table 8
Testing results of residential house merged orthophotos.

Model	Overall			Overall (excl. *background*)			*background*		*slate*		*leadwork*		*other*	
	aAcc	mIoU	mAcc	aAcc	mIoU	mAcc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Validation	95.22	65.79	89.77	89.45	54.39	86.37	100	100	88.86	89.48	49.59	91.99	24.72	77.63

roof contains 13 panels, all of them are planar. The photogrammetric reconstruction and the DT model of the roof are presented in Fig. 19.

4 representative panels are used for validating the orthophoto generation method proposed in Section 4: The results obtained for the generation of the orthophotos for the four panels are shown in Fig. 20. The $\bar{a}$ values for the four panels are: 13%, 24%, 6% and 14%, respectively. It can be seen that Panel S1 and S2 have lower $\bar{a}$ values, which again correlates with their larger size (width). Shadows of chimney in orange colour are also visible, implying that they have not been completely removed by the merging process. Panel S3 and S4 are better in quality with higher $\bar{a}$ values.

The orthophotos generated are then inputted in the trained model PointRend(CEL+FL) (from Section 7.5), for validation. The results are listed in Table 8.

The results for the tested orthophotos are shown in Fig. 20, alongside the orthophotos. The results are generally good. However, shadows from other components (e.g. chimney in this example) result in the deep learning models mistaking this area as *other*, which suggests some weakness in our merging strategy.

9. Discussion

9.1. Orthophoto generation

The proposed method for generating vertical orthophotos free of occlusions shows decent results. Occlusions are currently handled by analysing the frequency of RGB values and selecting the median value. While the method works reasonably well in the four representative examples presented, in some extreme cases, occlusions could still appear in a large(r) number of images, which would defeat the proposed approach and cause shadows in the final merged orthophotos. For example, when some decorative components are placed right above or too close to the target panel, only when the UAV is flown exactly at the right location can the entire panel be captured without occlusion, which would require great skills as well as meticulous planning by the operator. However, the incidence angle in such situation is usually quite oblique, which will be filtered out in the process of refinement described in Section 4.2.

An alternative, more robust approach to deal with occlusions is to use spatial information (e.g. depth information), to refine the image masks, and thereby filter out the pixels from the occluders. Indeed, if the panel view is unobstructed, the depth associated to a pixel should be the range of the ray going through the pixel and intersecting the

roof panel plane. But, if the ray intersects another building element, e.g. chimney (see Figs. 7, 8, 19), then the pixel should be marked as 'occluded' and added to the mask. The remaining steps of the proposed process would then remain unchanged.

9.2. Semantic segmentation

Despite the positive results obtained, the final segmentation results show that some segments are still not clean, including pixels with erroneous classification. This would possibly cause confusion to subsequent data analysis, such as defect detection. Therefore, methods should be applied to resolve these errors. This could include morphological transformation to replace the pixel label with the that of neighbourhood ones.

Besides, future work can look at collecting more building roof data in order to further validate and improve the orthophoto generation pipeline, and train semantic segmentation models with greater generalisability. These data shall include roofs with various forms and shapes, including slate laying methods and different components around the slated areas, and in various conditions, containing different levels of deteriorations. Another approach for semantic segmentation could also be explored, for example building on the latest segmentation tools like Segment Anything Model (SAM) [70].

Subsequent work should also look at the next step of our proposed overall pipeline, i.e. the detection of defects in the slated and leadwork areas. Due to the fact that all the created orthophotos have the same orientation and uniform scale, we anticipate that this should ease the development of machine learning models.

Finally, work can be conducted into the means to populate and maintain the roof's (and entire building's) Digital Twin or Digital Logbook with the extracted information, such as the semantic segmentation output, detected defects and their characteristics, so that building owners and maintenance contractors can make comprehensive assessment of the evolution of defects, conduct holistic diagnostics, make accurate calculation on how much repair materials are needed, etc.

10. Conclusions

A general pipeline for automatically monitoring individual panels of pitched building roof and maintaining digital twins (or digital logbooks) of building roofs, is proposed in this study. The advantages it demonstrates, especially its automation feature, can effectively address the challenges that traditional roof survey is facing as mentioned in Section 1: reducing time, cost, and safety hazards. It is composed of three stages: roof panel orthophoto generation, roof panel orthophoto semantic segmentation and deterioration and defect detection in the semantic segments. This paper then presents methods and validation results for the first two stages, which take two inputs: a structure-from-motion photogrammetric reconstruction, and the Digital Twin 3D model of the building being monitored.

A new orthophoto generation method is proposed that creates individual orthophotos for each building element, illustrated specifically in the challenging context of pitched building roofs. It outputs occlusion-free individual vertical orthogonal images specified with classes for each building element. To solve the problems existing in the current normal orthophoto generation approaches for entire façades, which are lacking semantic link to the as-built BIM model, and contain occlusions, the proposed method combines building reconstruction, camera projection, and orthophoto image generation with occlusion removal. The advantages of this method include:

1. The final orthophotos contain data for all parts of the roof elements and occlusions by other building elements are eliminated through the proposed merging mechanism that calculates the median orthophoto.
2. By introducing image filtering parameters (i.e. incidence angle and distances), the quality of the final merging orthophoto is further improved, with less blur and seemingly improved colour balancing.
3. Given a photogrammetric reconstruction and corresponding 3D DT model, the orthophoto generation process is fully automatic, and could bring convenience to current surveying practice, as well as emerging digitalised and automated methodologies, with the orthophotos (and any result that would be obtained from their analysis) semantically linked to elements in the Digital Twin or Digital Logbook — of the building.
4. The proposed method can be applied to drone survey data acquired (at frequent intervals in the case of pro-active maintenance schemes) over the life of the building. The orthophotos acquired at each epoch can be used to detect defects and the comparison of the results across epochs be used to assess the evolution of defects over time.

For the second stage, different deep learning models for semantic segmentation of the orthophoto images are developed and compared using a dataset composed of data coming from two buildings: Duff House, St Mary's Church. PointRend added with Focal Loss (PointRend(CEL+FL)), trained by the combined dataset is chosen as the most suitable when considering both datasets jointly. The contribution includes:

1. An accurate model for semantic segmentation is built.
2. A background label correction step is added. By eliminating the confusion between *background* and other classes (especially *leadwork* and *other*, which are usually the surrounding area of *slate*), the *background* accuracy is corrected to 100%, and the accuracies of other classes are shown to also increase.
3. Finally, two different strategies are compared so that only a single orthophoto with semantic segmentation results is outputted. The strategy of merging semantic segmentation results from unmerged orthophotos performs best. It not only improves the relative accuracy and pixel classification, but it also greatly eliminates confusions between the *slate* and *other* classes. Nonetheless, the confusion between *leadwork* and *other* remains relatively more significant, affecting the accuracy of *other*. This is possibly because that *other* class includes many kinds of objects. Enhancing the model robustness with more data can be a possible solution.

This study proposed a general solution, because it does not include any specific settings of the involved buildings. Therefore, as long as the building data is collected by photogrammetry, such pipeline can still be applied. From a broader perspective, this method can also be applied to other structures with a greater dimensions with difficulties that manual inspections are hard to carried out. Such structures include dams and bridges, and it is easy to fly a UAV above them. Besides, these buildings can provide more clear vision with less accessory components when flying a UAV, so that occlusion is less likely to happen.

CRediT authorship contribution statement

Jiajun Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Boan Tao:** Software. **Frédéric Bosché:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Chris Xiaoxuan Lu:** Writing – review & editing, Supervision, Methodology. **Lyn Wilson:** Writing – review & editing, Supervision, Resources, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jiajun Li reports financial support was provided by Engineering and Physical Sciences Research Council. Jiajun Li reports financial support was provided by Historic Environment Scotland. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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